

Directions(1-3): Answer the questions based on the information given below.

There are eight members in a family which consists of three generations and two married couples.

A is the brother-in-law of S, who is married. D is the granddaughter of N, who is a male. K is the father of R, who is the sister of S. B is the brother of A and the only son-in-law of M. M is the maternal grandmother of D.

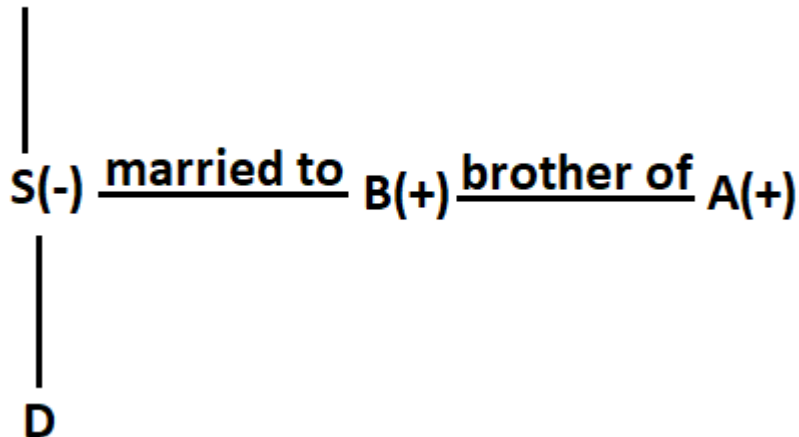
Common Solution:

Clues:

1. M is the maternal grandmother of D.
2. B is the brother of A and the only son-in-law of M.
3. A is the brother-in-law of S, who is married.

Inference: From clues 1, 2 and 3 we conclude that M is the mother of D's mother. B is married to S.

M (-)

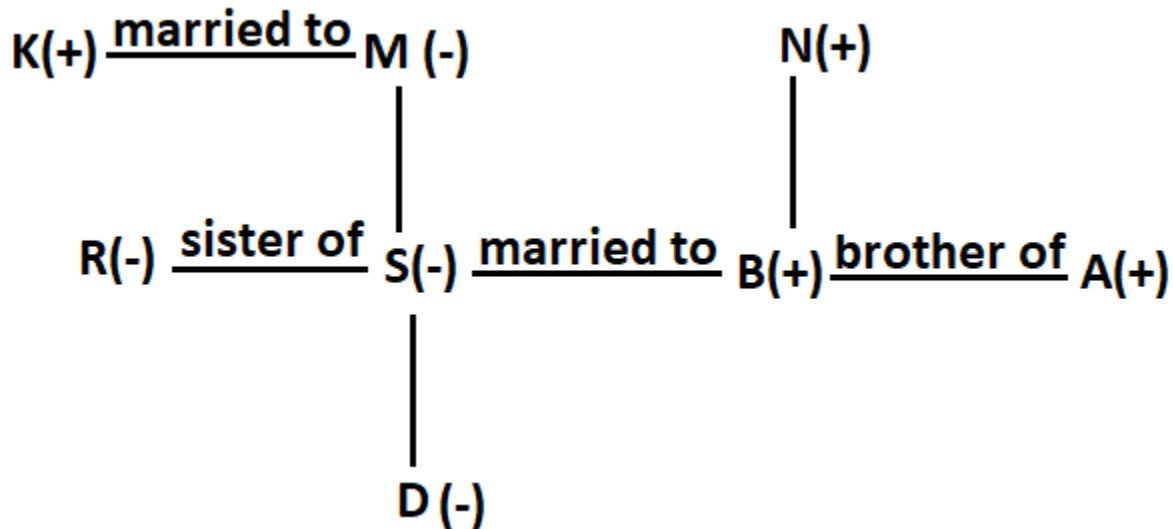


Clues:

4. K is the father of R, who is the sister of S.
5. D is the granddaughter of N, who is male.

Inference: From clues 4, 5, 6 and 7 we conclude that N is the paternal grandfather of D.

The family tree is given below:



Question 1: How is K related to B?

- A) Son
- B) Daughter
- C) Father-in law
- D) Daughter-in-law
- E) None of the above

Answer: C)

Solution:

K is the father-in-law of B.

Hence, option c.

Question 2: How is A related to N?

- A) Son-in-law
- B) Brother
- C) Son
- D) Nephew
- E) Can't be determined

Answer: C)

Solution:

A is the son of N.

Hence, option c.

Question 3: How is A related to D?

- A) Nephew
- B) Niece
- C) Aunt
- D) Uncle
- E) Can't be determined

Answer: D)

Solution:

A is the uncle of D

Hence, option d.

PRACTICE MOCK

Directions (4-8): Answer the questions based on the information given below.

Eight persons J, K, L, M, N, O, P and Q live on different floors of an eight-floor building such that the bottommost floor is numbered as 1 and the topmost floor is numbered as 8 but not necessarily in the same order. They work in eight different companies viz. HTC, Nike, Adidas, Spice, Dell, Nokia, HP, and Fila. One person works in only one company.

Q lives on an even numbered floor below the 6th floor. O lives three floors above Q. P works in Fila and lives on a floor above O. Three persons live between N and M. K and L works in Nike and Adidas respectively. Three persons live between P and J. N works in HTC and lives on an even numbered floor below P. K lives on one of the floors below L. J does not work in Spice, Dell or HP. Neither Q nor M works in Spice. M lives on an even numbered floor.

Common Solution:

Starting Point: Start with placing Q and O.

Clues:

1. Q lives on an even numbered floor below the 6th floor.
2. O live three floors above Q.
3. P works in Fila and lives on a floor above O.
4. Three persons live between P and J.

Inferences:

From clue 1, Q lives either on 4th or 2nd floor.

From clue 2, O lives either on 7th or 5th floor.

From clue 3, P lives on 8th, 6th, or 7th floor.

From clue 4, P lives on either 7th or 8th floor, J lives either on 3rd or 4th floor, Q lives on 2nd floor.

Case I: P lives on 7th floor.

Floors	Persons	Company
8		
7	P	Fila
6		
5	O	
4		
3	J	
2	Q	
1		

Case II: P lives on 8th floor.

Floors	Persons	Company
8	P	Fila
7		
6		
5	O	
4	J	
3		
2	Q	
1		

Clues:

- N works in HTC and lives on an even numbered floor below P.
- Three persons lives between N and M.
- M lives on an even numbered floor.
- K lives on one of the floors below L.

Inferences:

From clue 5, clue 6 and clue 7, N lives on 4th floor, Case II is rejected as N and M can't be placed.

From clue 7 and clue 8, M lives on 8th floor, K lives on 1st floor and L lives on 6th floor.

Floors	Persons	Company
8	M	
7	P	Fila
6	L	
5	O	
4	N	HTC
3	J	
2	Q	
1	K	

Clues:

9. K and L work in Nike and Adidas respectively.

10. J does not work in Spice, Dell or HP.

11. Neither Q nor M works in Spice.

Inferences:

From clue 10, J works in Nokia.

From clue 11, O works in Spice, either Q or M works in HP or Dell.

The final arrangement is as follows:

Floors	Persons	Company
8	M	HP/Dell
7	P	Fila
6	L	Adidas
5	O	Spice
4	N	HTC
3	J	Nokia

2	Q	Dell/HP
1	K	Nike

Question 4: Which of the following statements is/are definitely correct?

I. M works in Dell

II. Q works in HP

III. J works in Nokia

A) Only Statement I is correct

B) Only Statement II is correct

C) Only Statement III is correct

D) Only Statement I and II is correct

E) All Statements are correct

Answer: C)

Solution:

J works in Nokia is a correct statement.

Hence, option c.

Question 5: Who lives on 6th floor?

A) N

B) K

C) L

D) M

E) None of the above

Answer: C)

Solution:

L lives on 6th floor.

Hence, option c.

Question 6: How many person(s) lives between the one, who works in Nokia and the one, who works in Spice?

A) Three

B) One

C) Two

D) Four

E) None of the above

Answer: B)

Solution:

One person lives between the one, who works in Nokia and the one, who works in Spice

Hence, option b.

Question 7: M works in_____.

A) Dell

B) HP

C) Nokia

D) Either (a) or (b)

E) Either (b) or (c)

Answer: D)

Solution:

M works in either Dell or HP.

Hence, option d.

Question 8: Person, who lives on 1st floor, works in_____.

A) Adidas

B) Nike

C) Spice

D) Nokia

E) None of the above

Answer: B)

Solution:

Person, who lives on 1st floor, works in Nike.

Hence, option b.

Question 9: How many such pair(s) of letters are in the word “**ATTENDANT**” which has as many letters between them (both forward and backward) as in the English alphabetical series?

A) None

B) One

C) Two

- D) Three
- E) None of the above

Answer: B)

Solution:

Given word:

ATTENDANT

There is one such pairs (NT) which have as many letters between them (both forward and backward) as in the English alphabetical series.

Hence, option b.

Question 10: If we add one to all the digits of the number **2625684775** and then arrange them in ascending order from the left end, then which digit will be 6th from the left end after the rearrangement?

- A) 5
- B) 7
- C) 8
- D) 9
- E) None of the above

Answer: B)

Solution:

Given number:

2625684775

After adding one to all digits:

3736795886

After rearrangement in ascending order:

3356677889

The 6th digit from the left end after rearrangement is 7.

Hence, option b.

Directions(11-15): Answer the questions based on the information given below.

Seven persons A, B, C, D, E, F and G are sitting in Row 1 facing north whereas S, T, V, W, X, Y, and Z are sitting in Row 2 facing south such that both rows are equidistant and parallel to each other. The persons sitting in Row 1 face the persons sitting in Row 2 and vice versa.

D sits 4th to the right of F, who sits 2nd to the left of E. Either F or D sits at one of the extreme ends. W faces E. Y does not sit at any extreme ends of the row. Two persons sit between A and B. C does not face V. X does not face F or D. Two persons sit between W and S. V sits 4th to the right of S, who does not sit adjacent to Z. A faces Y.

Common Solution:

Starting Point: Start with placing D, F, and E.

Clues:

1. D sits 4th to the right of F, who sits 2nd to the left of E.
2. Either F or D sits at one of the extreme ends.
3. W faces E.
4. Two persons sit between W and S.

Inferences:

From clue 1 and clue 2, E sits 3rd from either of the extreme ends of the row.

From clue 3 and clue 4, S sits 3rd to the left or 3rd to the right of W.

Case I: When E sits 3rd from extreme left end of row.

		W			S	
F		E		D		

Case II: When E sits 3rd from extreme right end of row.

	S			W		
		F		E		D

Clues:

5. V sits 4th to the right of S, who does not sit adjacent to Z.

6. A faces Y.

7. Y does not sit at any extreme ends of the row.

8. Two persons sit between A and B.

Inferences:

From clue 5, case II is rejected as V can't be placed. V sits immediate right of W.

From clue 6 and clue 7, A sits immediate left of D.

From clue 8, B sits 2nd to the right of D.

	V	W	Y		S	
F		E	A	D		B

Clues:

9. C does not face V.

10. X does not face F or D.

Inferences:

From clue 9, C faces S. G faces V.

From clue 10, B faces X.

From clue 4, Z faces F. T faces D.

The final arrangement is as follows:

Z	V	W	Y	T	S	X
F	G	E	A	D	C	B

Question 11: What is the position of A with respect to B?

A) 2nd to the left

B) 2nd to the right

C) 3rd to the left

D) 3rd to the right

E) None of the above

Answer: C)

Solution:

A sits 3rd to the left of B.

Hence, option c.

Question 12: ____ sits 2nd to the left of Y.

A) Z

B) V

C) X

D) S

E) W

Answer: D)

Solution:

S sits 2nd to the left of Y.

Hence, option d.

Question 13: How many persons sit between Z and the one, who faces D?

A) Three

B) One

C) Two

D) None

E) Cannot be determined

Answer: A)

Solution:

Three persons sit between Z and T, who faces D.

Hence, option a.

Question 14: D sits _____ to the left of C.

A) Immediate

B) Second

C) Fourth

D) Third

E) None of the above

Answer: A)

Solution:

D sits immediate to the left of C.

Hence, option a.

Question 15: Which of the following statement is/are not correct?

A) V sits immediate left of Z

B) X sits immediate left of S

C) E sits 2nd to the left of F

D) E sits immediate right of G

E) None of the above

Answer: C)

Solution:

E sits 2nd to the left of F is an incorrect statement.

Hence, option c.

Question 16: In the question, assuming the given statements to be true, find which of the following conclusion(s) among the three conclusions is/are true and then give your answer accordingly.

Statement:

$I \geq J \geq R = U \geq S$; $T < M < N$; $L = S \geq T$

Conclusions:

I. $I \geq M$

II. $R \geq T$

III. $J > S$

A) Only conclusion III is true.

B) Only conclusion I and II are true.

C) Only conclusion II is true

D) Only conclusion II and III are true.

E) All the conclusions are true

Answer: C)

Solution:

Given:

$I \geq J \geq R = U \geq S$; $T < M < N$; $L = S \geq T$

On combining the given statement, we have

$I \geq J \geq R = U \geq L = S \geq T < M < N$

Conclusion:

I. $I \geq M$: False (As $I \geq J \geq R = U \geq L = S \geq T < M$, so relation between I and M cannot be determined)

II. $R \geq T$: True (As $R = U \geq L = S \geq T$, so $R \geq T$)

III. $J > S$: False (As $J \geq R = U \geq L = S$, so $J \geq S$)

Only conclusion II is true.

Hence, option c.

Question 17: In the question, assuming the given statements to be true, find which of the following conclusion(s) among the three conclusions is/are true and then give your answer accordingly.

Statement:

$J < R < G < L < M$; $N > T > M$; $J > K = C$

Conclusions:

I. $T < C$

II. $N < G$

III. $M > R$

A) Only conclusion III is true.

B) Only conclusion I and II are true.

C) Only conclusion II is true

D) Only conclusion II and III are true.

E) All the conclusions are true

Answer: A)

Solution:

Given:

$J < R < G < L < M$; $N > T > M$; $J > K = C$

On combining the given statement, we have,

$C = K < J < R < G < L < M < T < N$

Conclusion:

I. $T < C$: False (As $C = K < J < R < G < L < M < T$, so $T > C$)

II. $N < G$: False (As $G < L < M < T < N$, so $N > G$)

III. $M > R$: True (As $R < G < L < M$, so $M > R$)

Only conclusion III is true.

Hence, option a.

Question 18: In the question, assuming the given statements to be true, find which of the following conclusion(s) among the three conclusions is/are true and then give your answer accordingly.

Statement:

$P = R \geq O = M$, $S < E \leq O$, $T < K \leq R$

Conclusions:

I. $P > S$

II. $T \geq O$

III. $E \leq R$

A) Only conclusion III is true.

B) Only conclusion I and III are true.

C) Only conclusion II is true

D) Only conclusion II and III are true.

E) All the conclusions are true

Answer: B)

Solution:

Given:

$$P=R\geq O=M, S<E\leq O, T<K\leq R$$

On combining the given statement, we have,

$$T<K\leq P=R\geq O=M\geq E>S$$

Conclusion:

I. $P>S$: True (As $P=R\geq O=M\geq E>S$, so $P>S$)

II. $T\geq O$: False (As $T<K\leq P=R\geq O$, so relation between T and O cannot be determined)

III. $E\leq R$: True (As $R\geq O=M\geq E$, so, $E\leq R$)

Only conclusion I and III are true.

Hence, option b.

Question 19: Which of the following symbols should be placed in the blank spaces respectively (in the same order from left to right) in order to complete the given expression in such a manner that ' $R\geq Y$ ' definitely holds true?

A _ R _ T _ L _ Y

A) $>, <, \geq, =$

B) $>, >, >, =$

C) $<, >, \geq, =$

D) $>, \geq, \geq, =$

E) $<, >, \geq, \leq$

Answer: D)

Solution:

On substituting option in given expression:

Option (a): $A > R < T \geq L = Y$: Relation between R and Y can't be determined.

Option (b): $A > R > T > L = Y$, so $R > Y$.

Option (c): $A < R > T \geq L = Y$, so $R > Y$.

Option (d): $A > R \geq T \geq L = Y$, so $R \geq Y$

Option (e): $A < R > T \geq L \leq Y$: Relation between R and Y can't be determined.

Hence, option d.

Directions(20-23): Answer the questions based on the information given below.

Nine persons are posted in the Indian army at different designations. Designations (from senior to junior) are Admiral, General, Brigadier, Lieutenant, Colonel, Major, Captain, Subedar, and Havildar i.e. Admiral is the highest designation and Havildar is the lowest designation.

Only three persons are junior to T. Two designations lie between the designations of O and R, who is senior to O. R is neither Admiral nor colonel. M is senior to Q but junior to U. U is junior to N. U is not designated as a lieutenant. Three designations lie between the designations of S and P, who is junior to T. S is not designated as Colonel.

Common Solution:

Starting Point: First consider the direct statements/clues.

Clue:

1. Only three persons are junior to T.
2. Three persons are designated between S and P, who is junior to T.
3. S is not designated as Colonel.

Inference: From clues 1, 2 and 3 we conclude that there will be two possible ways to arrange S and P.

Case 1: When S was Lieutenant.

Designation	Person
Admiral	
General	
Brigadier	
Lieutenant	S
Colonel	
Major	T
Captain	
Subedar	P
Havildar	

Case 2: When S was Brigadier.

Designation	Person
Admiral	
General	
Brigadier	S
Lieutenant	
Colonel	
Major	T
Captain	P
Subedar	
Havildar	

Clue:

4. Two persons are designated between O and R, who was senior to O.

5. R was neither Admiral nor colonel.

Inference:

From clue 4 and clue 5, O is the Colonel. R is the General.

Case 1: When S was Lieutenant.

Designation	Person
Admiral	
General	R
Brigadier	
Lieutenant	S
Colonel	O
Major	T
Captain	
Subedar	P
Havildar	

Case 2: When S was Brigadier.

Designation	Person
Admiral	
General	R
Brigadier	S
Lieutenant	
Colonel	O
Major	T
Captain	P
Subedar	
Havildar	

Clues:

6. M is senior to Q but junior to U.

7. U is junior to N.

8. U is not designated as a lieutenant.

Inference: From clues 6, 7 and 8, case 2 is rejected. And thus, N is the Admiral, U is the Brigadier, Q is the havildar.

The final table is given below:

Designation	Person
Admiral	N
General	R
Brigadier	U
Lieutenant	S
Colonel	O
Major	T
Captain	M
Subedar	P
Havildar	Q

Question 20: Who among the following is the Admiral?

- A) S
- B) R
- C) U
- D) N
- E) M

Answer: D)

Solution:

N is the Admiral.

Hence, option d.

Question 21: How many persons are junior to O?

- A) Three
- B) None
- C) Two
- D) Five
- E) None of the above

Answer: E)

Solution:

Four persons are junior to O.

Hence, option e.

Question 22: What is the designation of P?

- A) Subedar
- B) Captain
- C) Colonel
- D) Major
- E) None of the above

Answer: A)

Solution:

P is designated as Subedar.

Hence, option a.

Question 23: How many persons are designated between N and Captain?

- A) Four

- B) Three
- C) Five
- D) Six
- E) None of the above

Answer: C)

Solution:

Five persons are designated between N and Captain.

Hence, option c.

Directions(24-28): Answer the following question based on the information given below.

In a certain code language,

- I. 'some birds fly high' is written as 'rop, lup, rts, mus'
- II. 'some fish swim deep' is written as 'rop, jhr, bjk, mok'.
- III. 'birds are high' is written as 'vot, rts, lup'.
- IV. 'high and deep' is written as 'bjk, lup, das'.

Common Solution:

From statement I, III and IV, 'high' is coded as 'lup'.

From statement I and II, 'some' is coded as 'rop'.

From statement I and III, 'birds' is coded as 'rts'.

From statement I alone, 'fly' is coded as 'mus'.

From statement II and IV, 'deep' is coded as 'bjk'.

From statement III alone, 'are' is coded as 'vot'.

From statement IV alone, 'and' is coded as 'das'.

From statement II, 'fish' and 'swim' is coded as 'jhr' and 'mok'.

Word	High	some	birds	fly	deep	and	are	fish	swim
Code	lup	rop	rts	mus	bjk	das	vot	mok/jhr	jhr/mok

Question 24: What is the code for 'deep'?

- A) jhr
- B) bjk
- C) mok
- D) das
- E) None of the above

Answer: B)

Solution:

The code for 'deep' is 'bjk'.

Hence, option b.

Question 25: Which of the following is coded as 'vot'?

- A) are
- B) bird
- C) high
- D) deep
- E) None of the above

Answer: A)

Solution:

‘are’ is coded as ‘vot’.

Hence, option a.

Question 26: What is the code for ‘fish and high’?

- A) mok das lup
- B) rts jhr mok
- C) mok vot das
- D) jhr das lup
- E) Cannot be determined

Answer: E)

Solution:

Code for ‘fish and high’ can be ‘mok das lup’ or ‘jhr das lup’.

Hence, option e.

Question 27: ____ is coded as ‘mus’.

- A) deep
- B) birds
- C) fly
- D) are
- E) None of the above

Answer: C)

Solution:

‘fly’ is coded as ‘mus’.

Hence, option c.

Question 28: Which of the following statement is incorrect?

- A) ‘and’ is coded as ‘das’.
- B) ‘bjk’ is the code for ‘deep’.
- C) ‘some birds’ is coded as ‘rop rts’.
- D) ‘some fish’ is coded as ‘mok jhr’
- E) All statements are incorrect.

Answer: D)

Solution:

Option (a): Correct (‘and’ is coded as ‘das’.)

Option (b): Correct (‘bjk’ is the code for ‘deep’.)

Option (c): Correct (‘some birds’ is coded as ‘rop rts’.)

Option (d): Incorrect (‘some fish’ is coded as ‘rop mok’ or ‘rop jhr’)

Hence, option d.

Directions(29-33): Answer the questions based on the information given below.

Eight persons (P, Q, R, S, T, U, V and W) were born in eight different years (1984, 1988, 1991, 1993, 1997, 2000, 2003 and 2008) but not necessarily in the same order.

Note: Consider all persons were born on 31st December in given years. Ages of these persons are calculated with respect to 31st December 2021.

S was born in an odd numbered year before Q. T is elder than S but not the eldest. V was born in one of the years before R and in a leap year. The age difference between W and P is 5 years. U was born in a leap year. Q is 10 years elder than W.

Common Solution:

Starting Point: First we calculate age of persons according to their birth year. (37, 33, 30, 28, 24, 21, 18 and 13) (in years) with respect to 31st December 2021. Now we start with definite clues/information.

Clues:

1. The age difference between W and P is 5 years.
2. Q is 10 years elder than W.

Inferences:

From clue 1 and 2, we get W was born in 2003, P was born in 2008, and Q was born in 1993.

Years	Ages	Persons
1984	37	
1988	33	
1991	30	
1993	28	Q
1997	24	
2000	21	
2003	18	W
2008	13	P

Clues:

3. S was born in an odd numbered year before Q.
4. T is elder than S but not the eldest.
5. V was born in one of the years before R and in a leap year.

6. U was born in a leap year.

Inferences:

From clue 3, we get S was born in 1991.

From clue 4, we get T was born in 1988.

From clue 5 and 6, V was born in 1984, R was born in 1997 and U was born in 2000.

The final arrangement is as follows:

Years	Ages	Persons
1984	37	V
1988	33	T
1991	30	S
1993	28	Q
1997	24	R
2000	21	U
2003	18	W
2008	13	P

Question 29: ____ was born in 1988.

- A) W
- B) P
- C) S
- D) T
- E) None of the above

Answer: D)

Solution:

T was born in 1988.

Hence, option d.

Question 30: V is ____ elder than S.

- A) 5 years
- B) 4 years
- C) 7 years
- D) 3 years
- E) None of the above

Answer: C)

Solution:

V is 7 years elder than S.

Hence, option c.

Question 31: How many persons are younger than U?

- A) Three
- B) One
- C) Two
- D) Five
- E) Four

Answer: C)

Solution:

Two persons are younger than U.

Hence, option c.

Question 32: What is the sum of ages of V and R?

- A) 58 years
- B) 51 years
- C) 61 years
- D) 67 years
- E) None of the above

Answer: C)

Solution:

The sum of ages of the V and R is $(37 + 24) = 61$ years.

Hence, option c.

Question 33: P was born in_____.

- A) 2003
- B) 2008
- C) 1988
- D) 1993
- E) None of the above

Answer: B)

Solution:

P was born in 2008.

Hence, option b.

Question 34: In the question below some statements are given followed by two conclusions I and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusion definitely follows from the given statements, disregarding commonly known facts.

Statements:

A few Pens are Pencils

No Erasers are Pens

Some Crayons are Erasers

Conclusions:

I. Some Crayons are not Pens

II. At least Some Pencils are not Erasers

A) Neither Conclusion I nor Conclusion II follows

B) Only Conclusion I follows

C) Only Conclusion II follows

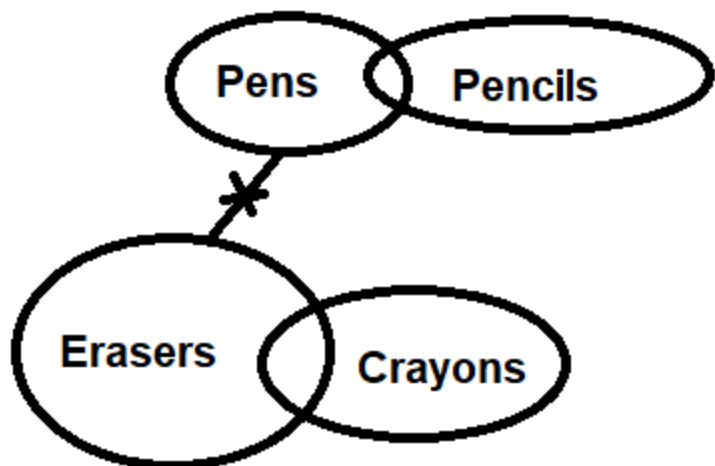
D) Both Conclusion I and Conclusion II follow

E) Either Conclusion I or Conclusion II follows

Answer: D)

Solution:

Following figure can be formed:



From the figure, both Conclusion I and Conclusion II follow.

Hence, option d.

Question 35: In the question below some statements are given followed by two conclusions I and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusion definitely follows from the given statements, disregarding commonly known facts.

Statements:

No Wells are Taps

Only Taps are Tubewells

Only a few Taps are Pipes

Conclusions:

I. No Pipes are Tubewells

II. Some Wells being Tubewells is a possibility

A) Neither Conclusion I nor Conclusion II follows

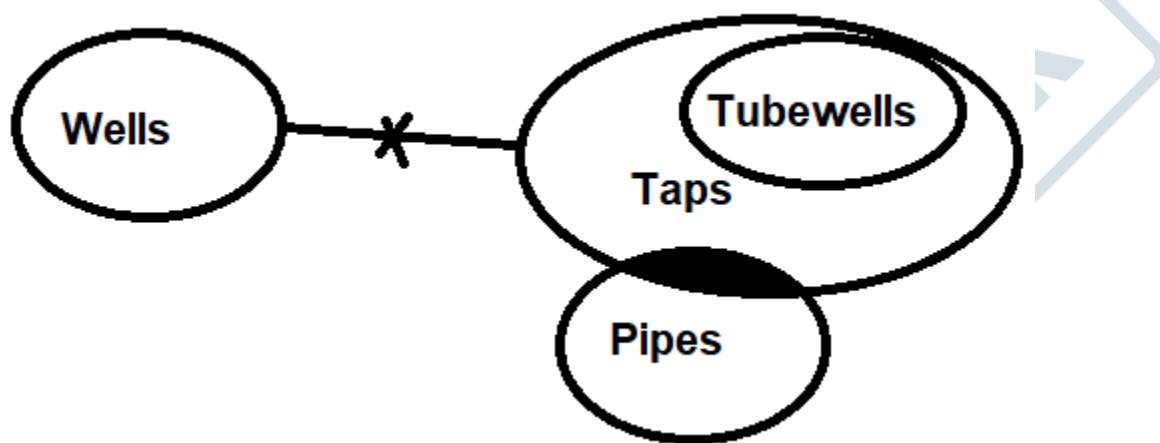
B) Only Conclusion II follows

- C) Only Conclusion I follows
- D) Both Conclusion I and Conclusion II follow
- E) Either Conclusion I or Conclusion II follows

Answer: C)

Solution:

Following figure can be formed:



From the figure, only Conclusion I follows.

Hence, option c.